

Hibernate Interview Questions and Answers

1. What is Hibernate?

Hibernate is an ORM (Object Relational Mapping) tool for Java. It maps Java classes to database tables.

2. Hibernate Object Lifecycle

1. Transient: Not associated with any Session.
2. Persistent: Associated with a Session and saved to DB.
3. Detached: Once persistent but Session is closed.
4. Removed: Object is marked for deletion from DB.

3. What is Session in Hibernate?

Session is used to get a physical connection with the database. It provides methods to create, read, and delete operations.

4. What is the difference between get() and load()?

- get(): Returns null if object not found.
- load(): Throws ObjectNotFoundException if object not found.

5. What are the advantages of Hibernate?

- Eliminates JDBC boilerplate code
- Caching support
- HQL
- Automatic table creation
- Lazy loading

6. What is caching in Hibernate?

- First-Level Cache: Session-scoped
- Second-Level Cache: SessionFactory scoped
- Query Cache: For query results

7. What are fetching strategies?

- Lazy: Loads associations only on access
- Eager: Loads associations immediately

8. What is HQL?

Hibernate Query Language, similar to SQL but works with entity names instead of table names.

9. Difference between save(), persist(), and saveOrUpdate()?

- save(): Returns ID, executes insert
- persist(): Doesn't return ID
- saveOrUpdate(): Inserts or updates record

10. What is the difference between merge() and update()?

- update(): Reattaches a detached object
- merge(): Copies state to a persistent instance